NOVA INFORMATION MANAGEMENT SCHOOL

Homework

Machine Learning
Data Science and Advanced Analytics

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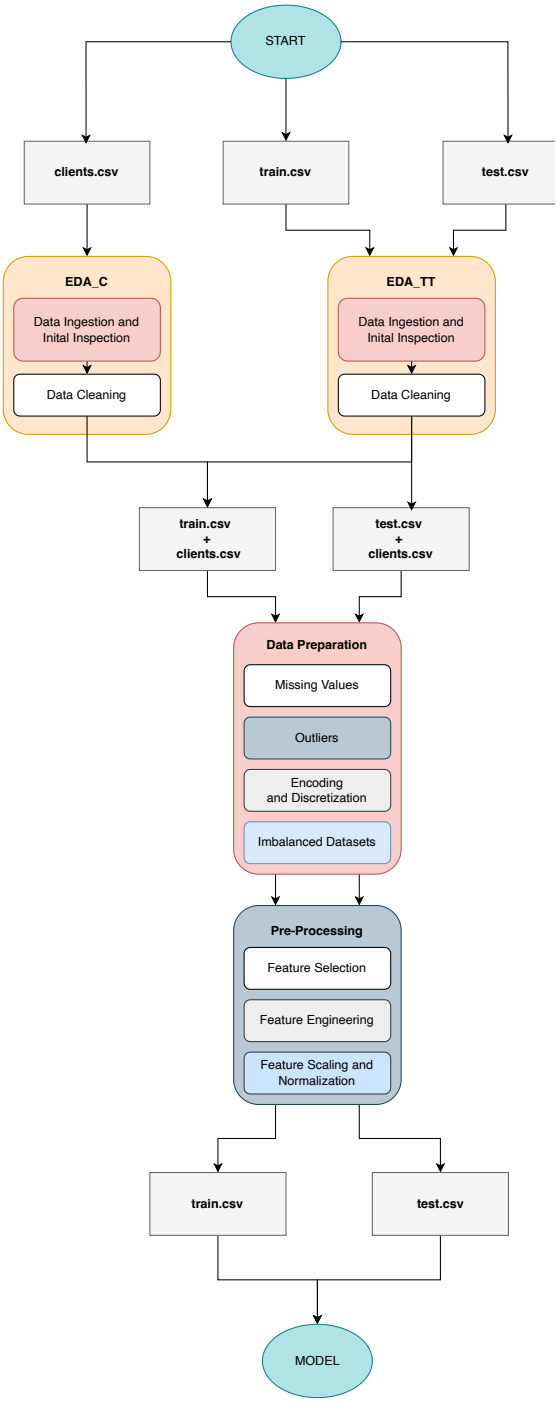
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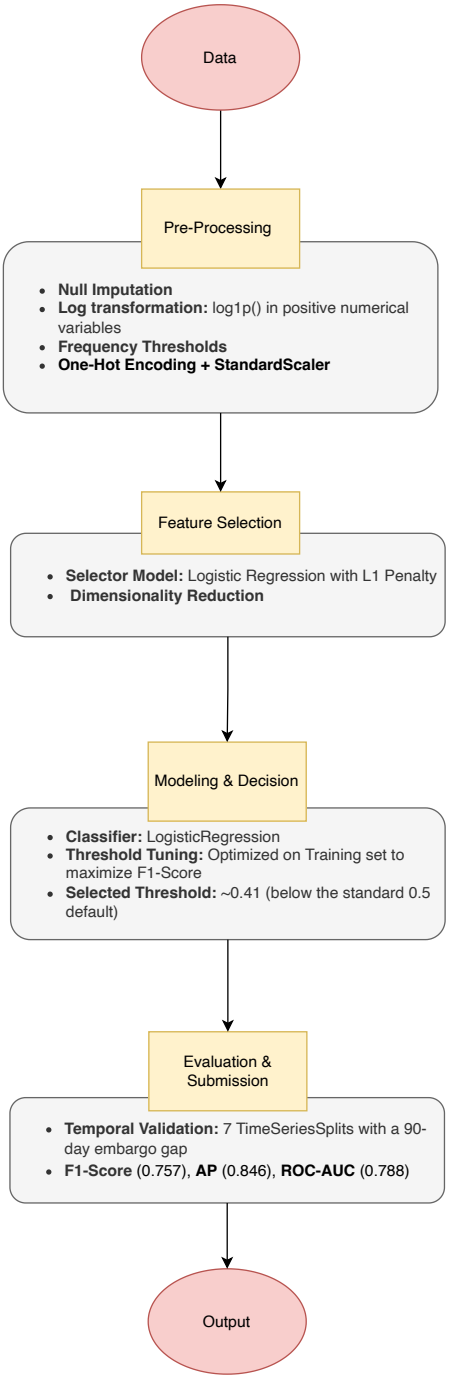
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1 Pipeline Structure



(a) Data Preparation Schema Pipeline.



(b) Model Schema Pipeline.

Figure 1: Pipeline Schemas.

1.1 Pre-Processing Summarization

The pipeline turns 6,534 raw purchases and a customer table into a logistic-regression submission; every step that learns from data is fitted on training rows only. Cleaning rules are fixed and applied once; the recipe, selection, model and cut-off are refitted inside each split, then once more on all training rows.

File	Rows × cols	Role
train.csv	6,534 × 39	Purchases 2015-07 to 2025-02; target repeat_purchase_90d (56% repeat)
test.csv	986 × 38	Purchases 2025-05 to 2026-10, no target
clientes.csv	1,143 × 13	One row per customer, joined on customer_id

Cleaning rules

- Drop 65 train rows without *Total a Pagar* (6,534 → 6,469) every test row is kept;
- Negatives in four columns that cannot be negative become gaps; first-time buyers get *is_first_purchase* (812 rows) and neutral fills;
- Inconsistencies repaired: *prior_mean_purchase_value* recomputed (272 cells), 15 spend_365d placeholders of 109 rebuilt, 45 prior_max values above total spend corrected;
- *purchase_date* becomes year, month, day, weekday, quarter and weekend flag.

Recipe (fitted on training rows only). Median/mode fill (six plans compared, none better; dropping incomplete rows was worse: AP 0.8106 vs 0.8326). Outliers kept (66.7% of rows have one; large buyers are real signal) and log1p on non-negative continuous columns. Rare levels → "Other"; one-hot with first level dropped; StandardScaler. ID, customer_id, company_name, address, street, postal_code and random_noise never enter the model.

1.2 Customer-Data Integration

clientes.csv joins every purchase on *customer_id* via a left join checked as many-to-one: all 6,469 training purchases (607 customers) find their customer, giving 57 columns. Customers are cleaned first: the empty customer_type is dropped; structural gaps in island, municipality and postal_area (Cramér's V 0.95–1.00) get the labels "(foreign)" / "(mainland)"; 2015 holds 58% of customers (records start July 2015), so it is kept with the flag since_2015_or_earlier. Identifiers are set aside; city, postal_area, municipality and transport_method were dropped in selection, raising mean F1 from 0.7268 to 0.7419. What survives: coarse geography (island, commercial_region) and customer_since_year.

1.3 Feature Selection

An L1-penalised logistic regression (C = 0.1), refitted inside each split, matches the 93 candidate columns while keeping 29 on average. Identifier-like columns (e.g. city: 259 levels, median 1 customer) were removed first. Techniques were compared by paired AP differences over 7 splits (significant only beyond two standard errors).

Technique	Family	Cols	AP / F1	Vs. all
All candidates	Reference	93	0.8456 / 0.7419	Ref.
Variance filter (0)	Filter	93	0.8456 / 0.7419	Identical
SelectKBest k=10	Filter	10	0.8317 / 0.7268	Worse
SelectKBest k=20	Filter	20	0.8464 / 0.7473	In noise
RFE k=20	Wrapper	20	0.8397 / 0.7327	Worse
L1 logistic C=0.1	Embedded	29	0.8459 / 0.7402	In noise

No technique improves prediction, so the goal is fewer columns at no cost. L1 was chosen because it judges columns jointly, keeps the same 17 columns in all 7 splits (RFE: 6), and its penalty was fixed in advance. On all training rows it keeps 39 of 106 encoded columns.

1.4 Modeling and Assessment Choices

Assessment. Random splits would leak the future (the target looks 90 days ahead, test is later in time and the repeat rate drifts 0.50–0.62). Purchases are date-ordered into 10 rolling blocks, each validated by a model trained on earlier purchases only; the 7 blocks with ≥ 2,000 training rows are kept, with a 90-day gap.

Metrics. AP decides comparisons (ranks returners without a cut-off); F1 is the competition metric but fragile at 0.5, so the final model uses the cut-off with best training F1: 0.41.

Model or floor	F1 (mean, 7 splits)
Predict a return for everyone	0.719
Baseline rule purchase_count_365d ≥ 3	0.746
Same model at 0.5 cut-off	0.740
Model at its cut-off (0.41)	**0.757 ± 0.011**

The model beats the rule by +0.010 ± 0.002 (ahead on 6 of 7 splits). Also: AP 0.846, ROC-AUC 0.788, balanced accuracy 0.702, precision 0.72, recall 0.79; training F1 0.763 shows no overfitting. The final model, refitted on all 6,469 rows, predicts 71.4% of the 986 test purchases as returning. Limitations: recipe, selection and cut-off were chosen on the same splits (slightly optimistic) and only one model family was assessed.

1.5 Step by Step Explanation

1. **Explore (1_EDA)**. No duplicates; 40% of train rows have a gap; purchase counts are the strongest signal (r = 0.33–0.34); train and test behave alike.
2. **Clean (1_EDA)**. Fixed rules for gaps, impossible values and identifiers, identical on train and test.
3. **Merge (2_Pre-processing)**. Checked left join on customer_id adds context without losing rows.
4. **Split in time**. Seven rolling-origin splits mimic predicting later purchases from earlier ones.
5. Preprocess per split. Fill, log1p, rare levels, one-hot, scaling – fitted on training rows only.
6. **Select (3_Feature_selection)**. Drop identifier-like columns, compare filter/wrapper/embedded, keep L1.
7. **Model (4_Model)**. Logistic regression scored with AP and F1 at a training-chosen cut-off, against two floors.
8. **Submit**. Refit on all 6,469 rows and predict the 986 test purchases, IDs unchanged.

Repository link: https://github.com/Diogosousa03/ML_Grupo18